

Capstone Project - Planning Document

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Project Overview

Website	snitch.com	Framework	Playwright v1.4x+ JavaScript Node.js
Domain	Fashion E-Commerce B2C	Pattern	Page Object Model (POM) + Custom Fixtures
Test Cases	120+ (8 modules x 15 each)	Reporting	Allure Report + Playwright HTML Report
AI Tools	GitHub Copilot + ChatGPT	CI/CD	GitHub Actions / Jenkins (headless execution)

Why snitch.com?

Feature-Rich Fashion E-Commerce	Modern Industry Standard UI	Complex Real-World Test Scenarios
Contains realistic e-commerce functionalities like filters, wishlist, product variants, dynamic product cards, and cart workflows.	Mimics a production-grade fashion storefront with responsive layouts, asynchronous loading, and interactive UI components.	Allows testing of real-world scenarios such as guest vs. registered user flows, coupon validations, responsive behaviors, and dynamic filtering systems.

Services & Test Case Breakdown (8 × 15 = 120 Test Cases)

#	Service / Module	Key Test Scenarios	Cases
1	Authentication	Valid login, invalid login, logout, registration flow validation, password field validation, session persistence, empty field validation, invalid email format handling, guest vs. logged-in states, password visibility toggle, redirection after login, authentication UI validation, social login button validation, browser back button behavior after login, remember me validation.	15
2	Product Catalog	Search by exact keyword, partial search, search with no results, category navigation, filter by size, filter by price, filter by color, sort by price low to high, sort by newest arrivals, pagination validation, breadcrumb navigation, product detail navigation, product image validation, out-of-stock product handling, product price accuracy.	15
3	Shopping Cart	Add single item, add multiple items, update product quantity, remove products, empty cart state validation, cart badge update,	15

		subtotal calculation accuracy, total price validation, continue shopping navigation, cart persistence after refresh, cart persistence after login, duplicate product handling, invalid quantity handling, coupon code validation, shipping estimation validation.	
4	Wishlist & Saved Items	Add item to wishlist, remove item from wishlist, wishlist persistence after login, move item from wishlist to cart, duplicate item handling, wishlist icon validation, empty wishlist state, add multiple products to wishlist, product variant handling in wishlist, responsive wishlist validation, product availability validation from wishlist, quick-view wishlist addition, navigation from wishlist to product page, clear wishlist validation, wishlist UI validation.	15
5	User Profile	Update first name and last name, update email validation, change password, address addition, edit existing address, delete address, required field validation, profile UI validation, order history access validation, newsletter subscription toggle, mobile responsiveness for profile pages, invalid input handling, profile navigation validation, session timeout handling, saved address validation.	15
6	Checkout Process	Guest checkout flow, logged-in checkout flow, shipping address validation, billing address validation, coupon application validation, invalid coupon handling, shipping method selection, payment page validation, order summary validation, tax calculation validation, empty cart checkout restriction, checkout form required field validation, navigation between checkout steps, responsive checkout validation, order confirmation page validation.	15
7	Customer Support & UI	Contact Us form submission, required field validation, invalid email handling, footer link validation, navbar validation, newsletter subscription validation, social media icon validation, responsive UI checks (mobile/tablet), banner validation, broken link verification, loading spinner validation, page redirection validation, FAQ navigation validation, browser compatibility validation, UI consistency validation.	15
8	E2E & Network Mocking	Complete End-to-End purchase flow, login to checkout workflow, visual regression on homepage, API response interception, product API mocking, forcing out-of-stock response validation, cart request interception, backend failure simulation (500 errors), timeout handling validation, slow network simulation, screenshot capture on failure, trace viewer generation, video recording validation, cross-browser E2E execution, parallel execution validation.	15

Implementation Milestones

Milestone	Deliverable
M1 - Setup & Config	Node.js workspace initialized, playwright.config.js configured with cross-browser profiles (Chromium, Firefox, WebKit), package.json scripts configured (test:headed, test:ui, test:report, test:allure).
M2 - Base POM & Locators	Core Page Object classes created (LoginPage, HomePage, ProductPage, CartPage). Robust locators implemented using getByRole(), getByText(), and stable selectors. Test data maintained using JSON files.
M3 - Advanced UI & Actions	Handling dynamic filters, hover interactions on navigation menus, cart table manipulations, responsive layouts, browser dialogs, file uploads, and multi-tab/window handling.
M4 - E2E & Assertions	Linking Page Objects to build complete End-to-End workflows (Login → Search → Add to Cart → Checkout). Implementation of soft assertions for non-critical UI validations.
M5 - Reporting & CI/CD	Integration of Allure Reports with screenshot, trace, and video attachments. Playwright HTML Reports configured. Failure artifact generation enabled. Basic GitHub Actions/Jenkins pipeline configured for headless execution.
M6 - Mocking & Final Polish	Using page.route() to intercept and mock backend API responses for edge-case testing. Final framework refactoring, reusable utility optimization, JSDoc comments, and README documentation completion.

Additional Framework Features

- Cross-browser testing support
- Parallel execution support
- Retry mechanism for flaky tests
- Screenshot capture on failure
- Trace and video recording
- Environment-based configuration handling
- Data-driven testing support
- Reusable utility methods
- Responsive viewport validation
- Structured reporting and debugging support

Website URL

<https://www.snitch.com/>